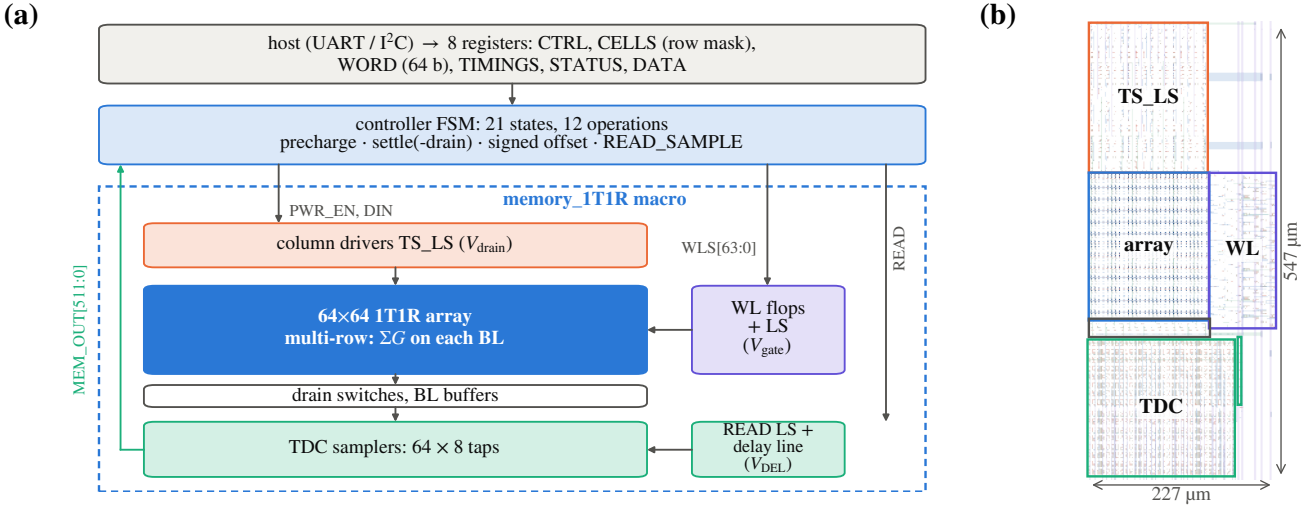
(a)
host (UART / I²C) → 8 registers: CTRL, CELLS (row mask),
WORD (64 b), TIMINGS, STATUS, DATA
controller FSM: 21 states, 12 operations
precharge · settle(-drain) · signed offset · READ_SAMPLE
memory_1T1R macro
PWR_EN, DIN
column drivers TS_LS (V_drain)
WLS[63:0]
READ
64×64 1T1R array
multi-row: ΣG on each BL
WL flops
+ LS
(V_gate)
drain switches, BL buffers
TDC samplers: 64 × 8 taps
READ LS +
delay line
(V_DEL)
MEM_OUT[511:0]
(b)
TS_LS
array
WL
TDC
547 µm
227 µm